Subalgebras of semisimple Lie algebras

Andrew Douglas

The City University of New York

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- The semisimple subalgebras of semisimple Lie algebras have received considerable attention.
  - Semisimple subalgebras of the exceptional Lie algebras, up to inner automorphism [Dynkin (1952), Minchenko (2006)].
  - Semisimple subalgebras of the simple Lie algebras of ranks $\leqslant 8$, up to linear equivalence [de Graaf (2009, 2011)].
- Less is known about nonsemisimple subalgebras of semisimple Lie algebras.
- In this talk, I will outline joint work with Repka, Kahrobaei, and de Guise on the classification of non-semisimple subalgebras of semisimple Lie algebras.
- Next, I will describe an application of the above work to a problem in physics.

- Unless otherwise stated, Lie algebras in this talk are finite-dimensional, over $\mathbb{C}$.

- Let $\varphi : \mathfrak{g}' \hookrightarrow \mathfrak{g}$ and $\varrho : \mathfrak{g}' \hookrightarrow \mathfrak{g}$ be Lie algebra embeddings.
- If φ and ϱ are equivalent up to inner automorphism we write $\varphi \sim \varrho$.
- φ and ϱ are linearly equivalent if for each representation $\pi : \mathfrak{g} \to \mathfrak{gl}(V)$ the induced $\mathfrak{g}'$-representations $\pi \circ \varphi$, $\pi \circ \varrho$ are equivalent, and we write $\varphi \sim_L \varrho$.

We define linear equivalence/equivalence of subalgebras much as we did for embeddings.

- Two subalgebras S and S' of $\mathfrak{g}$ are linearly equivalent if for every representation $\pi : \mathfrak{g} \to \mathfrak{gl}(V)$ the subalgebras $\pi(S)$, $\pi(S')$ of $\mathfrak{gl}(V)$ are conjugate under $GL(V)$.
- Two subalgebras S and S' of $\mathfrak{g}$ are equivalent if there is an inner automorphism ψ of $\mathfrak{g}$ such that $\psi(S) = S'$. In this case we write $S \sim S'$.
- Clearly if two embeddings or subalgebras are equivalent up to inner automorphism, then they are linearly equivalent. But, the converse is not in general true.

Levi decomposition

By Levi's Theorem (1905), any finite-dimensional Lie algebra $\mathfrak{g}$ over $\mathbb{C}$ may be decomposed as a semidirect sum

$$\mathfrak{g} = \mathfrak{p} \ltimes \mathfrak{r}, \ [\mathfrak{p}, \mathfrak{p}] = \mathfrak{p}, \ [\mathfrak{r}, \mathfrak{r}] \subsetneq \mathfrak{r}, \ [\mathfrak{p}, \mathfrak{r}] \subseteq \mathfrak{r}$$

where $\mathfrak{p}$ is semisimple, called the Levi factor, and $\mathfrak{r}$ is the radical of $\mathfrak{g}$. A Lie algebra is called a Levi decomposable algebra if its Levi decomposition is nontrivial.

Thanks to Rand, Winternitz, and Zassenhaus (1988) there exists an algorithm for realizing the Levi decomposition.

The Levi factor is unique up to isomorphism of $\mathfrak{g}$, and $\mathfrak{r}$ is unique.

Levi decomposable algebras in applications

Levi decomposable occur in many physical applications:

- The (complexification of the) Poincaré algebra $\mathfrak{p}(3,1) \cong \mathfrak{so}(4,\mathbb{C}) \ltimes (\mathbb{C}^2 \otimes \mathbb{C}^2)$ is the Lie algebra of isometries of Minkowski space-time.
- The Euclidean algebra $\mathfrak{e}(3) \cong \mathfrak{sl}(2,\mathbb{C}) \ltimes \mathbb{C}^3$ is the Lie algebra of orientation-preserving isometries of three-dimensional Euclidean space
- Levi decomposable algebras often occur as symmetry algebras of physically interesting PDEs.
- Levi decomposable algebras also occur in the construction of higher dimensional cosmological models as the algebras of Killing vectors.

Classification of Lie algebras

- No complete classification of Lie algebras exists, and is generally thought to be an impossible task.
- The task is facilitated by Levi's theorem.

By Levi's Theorem, classification of Lie algebras may proceed by examining

- semisimple Lie algebras
- solvable Lie algebras
- Levi decomposable algebras

Classification of semisimple Lie algebras

Semisimple Lie algebras over the field of complex numbers $\mathbb{C}$ have been completely classified by E. Cartan (1894).

- $A_n = \mathfrak{sl}(n+1, \mathbb{C}), n \geqslant 1$
- $B_n = \mathfrak{so}(2n+1, \mathbb{C}), n \geqslant 2$
- $C_n = \mathfrak{sp}(2n, \mathbb{C}), n \geqslant 3$
- $D_n = \mathfrak{so}(2n, \mathbb{C}), n \geqslant 4$
- $E_6,\ E_7,\ E_8$
- F_4
- G_2

Classification of solvable Lie algebras

- No classification of solvable Lie algebras exits, and such a classification is generally thought to be an impossible task.
- The problem stems from the fact that the number of solvable Lie algebras in higher dimensions increases drastically, and infinite parametrized families of such nonisomorphic algebras arise even in low dimensions.
- Special cases have been considered.
- There are two ways of proceeding in the classification of solvable Lie algebras: by dimension, or by structure.

Classification of solvable Lie algebras: Dimension

- The classification of low-dimensional solvable Lie algebras over $\mathbb{C}$ was started by S. Lie (1893) himself.
- Some incomplete classifications are known for solvable Lie algebras in dimension 7 and nilpotent algebras up to dimension 8 (M.P. Gong, Gr. Tsagas, A.R. Parry).
- W. de Graaf (2005) classified the solvable Lie algebras in dimension $\leqslant 4$ over fields of any characteristic.

Classification of solvable Lie algebras: Dimension

We describe de Graaf's classification over $\mathbb{C}$ of dimension 3 (dimension 4 becomes quite complicated):

J	The abelian Lie algebra of dimension 1
K_1	The abelian Lie algebra of dimension 2
K_2	$[z_1, z_2] = z_1$
L_1	The abelian Lie algebra of dimension 3
L_2	$[z_3, z_1] = z_1, [z_3, z_2] = z_2$
$L_{3,a}$	$[z_3, z_1] = z_2, [z_3, z_2] = az_1 + z_2$
L_4	$[z_3, z_1] = z_2, [z_3, z_2] = z_1$
L_5	$[z_3, z_1] = z_2$

Note that we get a nonisomorphic solvable Lie algebra $L_{3,a}$ for each $a \in \mathbb{C}$.

Classification of solvable Lie algebras

- It does not seem feasible to proceed by dimension in the classification of solvable Lie algebras beyond 6 dimensions.
- However, it is possible to proceed by structure. Specifically, classifying solvable Lie algebras with a nilradical of a given type has been fruitful.

Classification of solvable Lie algebras: Structure

- Heisenberg nilradicals: Rubin and Winternitz (1993),
- Abelian nilradicals: Ndogmo and Winternitz (1994),
- nilradicals of strictly upper triangular matrices: Tremblay and Winternitz (1998),
- two classes of filiform nilradicals: Snobl and Winternitz (2005),
- nilradicals with max. nilindex and Heisenberg subalgebra of codim. one: Ancochea, Campoamor-Stursberg, Vergnolle (2006),
- a certain sequence of quasi-filiform decomposable nilradicals: Wang, Lin, and Deng (2008).

Classification of Levi decomposable algebras

Given the semidirect sum $\mathfrak{p} \in_{\varphi} \mathfrak{r}$

- the Jacobi identity implies that $\mathfrak{r}$ is a representation space of $\mathfrak{p}$, and
- $\mathfrak{p}$ is isomorphic to some semisimple subalgebra of the algebra of derivations of $\mathfrak{r}$ (if $\varphi \neq 0$).
- This puts significant restrictions on which solvable algebras may occur in a Levi decomposable algebra.
- The Levi decomposable algebras up to dimension 9 were classified by Turkowski (1988, 1992).
- Levi decomposable algebras with nilpotent radicals that are of "low" nilpotent index have recently been classified by Benito and de-la-Conception (2013).

Subalgebras

By Levi's Theorem, classification of subalgebras of semisimple Lie algebras may proceed by examining the semisimple, solvable, and Levi decomposable subalgebras.

- Semisimple subalgebras have been studied extensively. Important examples have been examined, and more general techniques and algorithms have been developed: Dynkin, Minchenko, Lorente, Gruber, de Graaf.
- Non-semisimple (solvable and Levi decomposable) subalgebras have received less attention:
- D., Repka, Kahrobaei, de Guise have considered important examples, and developed more general techniques and algorithms.

Semisimple subalgebras

- Dynkin (1957) classified the simple subalgebras, up to linear equivalence, of the Lie algebras of exceptional type.
- Lorente and Gruber (1972) applied Dynkin's methods to classify the simple subalgebras of Lie algebras of classical type of ranks $\leqslant 6$, up to linear equivalence.
- Maltsev (1944) and Losev (2010) developed important techniques and theorems. For instance, their work gives us the following:
  - Equivalence implies linear equivalence, but the converse is not always true. However, if $\mathfrak{g}$ is of type A_n, B_n, C_n, F_4, G_2 then the two concepts coincide.

Semisimple subalgebras

- More recently Minchenko (2006) revisited Dynkin's classification of the simple subalgebras of the simple Lie algebras of exceptional type.
- Corrected a few small mistakes of Dynkin, and completed the classification of semisimple subalgebras of the exceptional Lie algebras, up to equivalence.
- Developed important theorems and techniques. The following is a generalization of a theorem by Dynkin:

Theorem

Let $\mathfrak{g}$ be a simple Lie algebra, and $\varphi_i : h \hookrightarrow \mathfrak{g}$, $i = 1, 2$, two embeddings.

1. *If $\mathfrak{g} \neq D_n$, then $\varphi_1 \sim_L \varphi_2$ iff $\pi_1 \circ \varphi_1 \sim \pi_1 \circ \varphi_2$;*
2. *If $\mathfrak{g} = D_n$, then $\varphi_1 \sim_L \varphi_2$ iff $\pi_k \circ \varphi_1 \sim \pi_k \circ \varphi_2$, $k = 1, n$,*

where π_i is the fundamental representation corresponding to the ith simple root.

Semisimple subalgebras

- De Graaf (2011) classified the semisimple subalgebras of the simple Lie algebras of ranks $\leqslant 8$ (linear equivalence).

	# linear equivalence classes
A_7	131
B_7	849
C_7	822
D_7	511
E_7	501
A_8	232
B_8	2186
C_8	2127
D_8	1664
E_8	1183

Non-semisimple subalgebras

Example 1. Abelian extensions of D_n into E_{n+1}

- D., Kahrobaei, and Repka (2013)
- An abelian extension of the special orthogonal Lie algebra D_n is a nonsemisimple Lie algebra $D_n \ltimes V$, where V is a finite-dimensional D_n-rep, and $[V, V] = 0$.
- We determined all abelian extensions of D_n that may be embedded into the exceptional Lie algebra E_{n+1}, $n = 5, 6$, and 7.
- We then classified these embeddings, up to inner automorphism.
- As an application, we also considered the restrictions of E_{n+1}-irreps to $D_n \ltimes V$, and discussed which of these restrictions are or are not indecomposable.

Non-semisimple subalgebras

Example 2. The Poincaré algebra $\mathfrak{p}(3,1)$.

- D., Repka, and de Guise (2013).
- The Poincaré algebra $\mathfrak{p}(3,1)$ is the Lie algebra of isometries of Minkowski space-time.
- $\mathfrak{so}(4,\mathbb{C}) \in (\mathbb{C}^2 \otimes \mathbb{C}^2)$ (Levi decomposable)
- We classified the Poincaré algebra in rank the 3 simple Lie algebras, up to inner automorophism.
- As an application, we also showed that each irreducible highest weight module of A_3 (not necessarily finite-dimensional) remains indecomposable when restricted to $\mathfrak{p}(3,1)$, with respect to any embedding of $\mathfrak{p}(3,1)$ into A_3.

Non-semisimple subalgebras

Example 3. Levi decomposable algebras in the classical Lie algebras.

- D. and Repka (2015).
- We classified the embeddings of $A_n \Subset \mathbb{C}^{n+1}$, $B_n \Subset \mathbb{C}^{2n+1}$, $C_n \Subset \mathbb{C}^{2n}$, and $D_n \Subset \mathbb{C}^{2n}$ into the complex simple Lie algebras A_{n+1}, B_{n+1}, C_{n+1}, and D_{n+1}, respectively, up to inner automorphism.

Subalgebras of A_2

- A classification of the semisimple subalgebras of A_2 is well-known.
- I will describe our classification of nonsemisimple subalgebras, which completes the classification of the subalgebras of A_2 [D., and Repka (2015)].

Chevalley basis of A_2

$$x_1 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad x_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad x_3 = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$y_1 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad y_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad y_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix},$$

$$h_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad h_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Semisimple subalgebras of A_2

There are precisely two semisimple subalgebras of A_2 up to linear equivalence, both isomorphic to the special linear algebra A_1. This is also true up to inner automorphism. They are

$$\begin{aligned} A_1^1 &\equiv \langle x_3, y_3, h_1 + h_2 \rangle, \\ A_1^2 &\equiv \langle x_1 + x_2, 2y_1 + 2y_2, 2h_1 + 2h_2 \rangle. \end{aligned}$$

Levi decomposable subalgebras of A_2

- We first decompose A_2 with respect to the adjoint actions of A_1^1 and A_1^2:

$$\begin{array}{lcccccccc} A_2 & \cong_{A_1^1} & \langle x_3, y_3, h_1 + h_2 \rangle & \oplus & \langle x_2, y_1 \rangle & \oplus & \langle x_1, y_2 \rangle & \oplus & \langle h_1 - h_2 \rangle \\ & \cong_{A_1^1} & V(2) & \oplus & V(1) & \oplus & V(1) & \oplus & V(0), \end{array}$$

$$\begin{array}{lcccc} A_2 & \cong_{A_1^2} & \langle x_1 + x_2, 2y_1 + 2y_2, 2h_1 + 2h_2 \rangle & \oplus & \langle x_3, x_1 - x_2, h_1 - h_2, y_1 - y_2, y_3 \rangle \\ & \cong_{A_1^2} & V(2) & \oplus & V(4). \end{array}$$

- The components–or combinations thereof–in the decompositions above, beyond A_1^1 and A_1^2, respectively, give us the possible extensions of A_1^1 and A_1^2, respectively.
- Once we've identified a list of (pairwise inequivalent) Levi decomposable subalgebras $A_1 \Subset_\varphi \mathfrak{r}$, we then must identify the radical in terms of de Graaf's classification, and the Lie algebra homomorphism $\varphi : A_1 \to Der(\mathfrak{r})$.

Levi decomposable subalgebras of A_2

$$
\begin{aligned}
\langle x_3, y_3, h_1 + h_2 \rangle \oplus \langle h_1 - h_2 \rangle &\cong A_1 \oplus J, \\
\langle x_3, y_3, h_1 + h_2 \rangle \Subset \langle x_1, y_2 \rangle &\cong A_1 \Subset_{\varphi_1} K_1, \\
\langle x_3, y_3, h_1 + h_2 \rangle \Subset \langle x_2, y_1 \rangle &\cong A_1 \Subset_{\varphi_1} K_1, \\
\langle x_3, y_3, h_1 + h_2 \rangle \Subset \langle h_1 - h_2, x_1, y_2 \rangle &\cong A_1 \Subset_{\varphi_2} L_2, \\
\langle x_3, y_3, h_1 + h_2 \rangle \Subset \langle h_1 - h_2, x_2, y_1 \rangle &\cong A_1 \Subset_{\varphi_2} L_2,
\end{aligned}
$$

$$
\left(\begin{array}{ll}
J & \text{The abelian Lie algebra of dimension 1} \\
K_1 & \text{The abelian Lie algebra of dimension 2} \\
L_2 & [z_3, z_1] = z_1, [z_3, z_2] = z_2
\end{array} \right)
$$

Solvable subalgebras of A_2

- We proceed by dimension to classify the solvable subalgebras of A_2.
- Let T be a solvable Lie algebra. Then there is a subalgebra $S \subset T$ of codimension 1, and $z \in T$ such that $T = \langle z \rangle \ltimes S$.
- The classification is with respect to de Graaf's classification of solvable Lie algebras.

One-dimensional subalgebras of A_2

- Classification of 1-dimensional subalgebras, up to inner automorphism, rests on the Jordan Canonical Form.

$$\begin{aligned} J^1 &= \langle x_1 + x_2 \rangle, \\ J^2 &= \langle x_1 \rangle, \\ J^3 &= \langle h_1 + 2h_2 + x_1 \rangle, \\ J^{4,\alpha} &= \langle h_1 + \alpha h_2 \rangle, \end{aligned}$$

for $\alpha \in \mathbb{C}$, where $J^{4,\alpha} \sim J^{4,\beta}$ if and only if $\beta = \alpha, \frac{1}{\alpha}, 1-\alpha, \frac{1}{1-\alpha}, \frac{\alpha}{-1+\alpha}$, or $\frac{-1+\alpha}{\alpha}$.

Two-dimensional subalgebras of A_2

- There are two possible 2-dimensional Lie algebras to consider: K_1 and K_2.
- A 2-dimensional subalgebra is constructed from the 1-dimensional subalgebras determined previously.

Two-dimensional subalgebras of A_2

$$
\begin{aligned}
K_1^1 &= \langle x_1 + x_2, x_3 \rangle, \\
K_1^2 &= \langle x_1, h_1 + 2h_2 \rangle, \\
K_1^3 &= \langle x_1, x_3 \rangle, \\
K_1^4 &= \langle x_1, y_2 \rangle, \\
K_1^5 &= \langle h_1, h_2 \rangle.
\end{aligned}
$$

$$
\begin{aligned}
K_2^1 &= \langle x_1 + x_2, h_1 + h_2 \rangle, \\
K_2^2 &= \langle x_1, -\tfrac{1}{3}h_2 + \tfrac{1}{3}h_2 + x_3 \rangle, \\
K_2^3 &= \langle x_1, -\tfrac{2}{3}h_1 - \tfrac{1}{3}h_2 + y_2 \rangle, \\
K_2^{4,\alpha} &= \langle x_1, \alpha h_1 + (2\alpha + 1)h_2 \rangle,
\end{aligned}
$$

where $K_2^{4,\alpha} \sim K_2^{4,\beta}$ if and only if $\alpha = \beta$.

Three-dimensional, solvable subalgebras of A_2

- By de Graaf's classification, a 3-dimensional, solvable subalgebra is isomorphic to one of the following:

L_1	The abelian Lie algebra of dimension 3
L_2	$[z_3, z_1] = z_1, [z_3, z_2] = z_2$
$L_{3,a}$	$[z_3, z_1] = z_2, [z_3, z_2] = az_1 + z_2$
L_4	$[z_3, z_1] = z_2, [z_3, z_2] = z_1$
L_5	$[z_3, z_1] = z_2$

- A 3-dimensional subalgebra is constructed from the 2-dimensional subalgebras determined previously.

Three-dimensional, solvable subalgebras of A_2

$$
\begin{aligned}
L_2^1 &= \langle x_1, x_3, 2h_1 + h_2 \rangle, \\
L_2^2 &= \langle x_1, y_2, h_1 - h_2 \rangle, \\
L_{3,-\frac{1}{4}}^1 &= \langle x_1, x_3, 2h_1 + h_2 + x_2 \rangle, \\
L_{3,-\frac{1}{4}}^2 &= \langle y_1, y_3, 2h_1 + h_2 + x_2 \rangle, \\
L_{3,-\frac{2}{9}}^1 &= \langle x_1 + x_2, x_3, h_1 + h_2 \rangle, \\
L_{3,0}^1 &= \langle x_1, h_1, h_2 \rangle, \\
L_{3,-\frac{(2\alpha-1)(\alpha-2)}{9(\alpha-1)^2}}^{1,\alpha} &= \langle x_1, x_3, (\alpha-1)h_1 + \alpha h_2 \rangle, \\
L_{3,-\frac{(2\alpha-1)(\alpha-2)}{9(\alpha-1)^2}}^{2,\alpha} &= \langle x_1, y_2, h_1 + \alpha h_2 \rangle, \\
L_4^1 &= \langle x_1, x_3, h_2 \rangle, \\
L_4^2 &= \langle x_1, y_2, h_1 + h_2 \rangle, \\
L_5^1 &= \langle x_1, x_2, x_3 \rangle,
\end{aligned}
$$

for $\alpha \neq \pm 1$.

Four-dimensional, solvable subalgebras of A_2

$$
\begin{aligned}
M_8^1 &= \langle x_1, x_3, h_1, h_2 \rangle, \\
M_8^2 &= \langle x_1, y_2, h_1, h_2 \rangle, \\
M_{12}^1 &= \langle x_1, x_2, x_3, h_1 + h_2 \rangle, \\
M^{1,\alpha}_{13,\frac{(2\alpha-1)(\alpha-2)}{(\alpha+1)^2}} &= \langle x_1, x_2, x_3, \alpha h_1 + h_2 \rangle, \\
M^2_{13,0} &= \langle x_2, y_1, y_3, 2h_1 + h_2 \rangle, \\
M^2_{13,2} &= \langle x_1, x_2, x_3, h_1 \rangle, \\
M^1_{14} &= \langle x_1, x_2, x_3, h_1 - h_2 \rangle,
\end{aligned}
$$

where $\alpha \neq \pm 1$; and $M^{1,\alpha}_{13,\frac{(2\alpha-1)(\alpha-2)}{(\alpha+1)^2}}$ and $M^{1,\beta}_{13,\frac{(2\beta-1)(\beta-2)}{(\beta+1)^2}}$ are equivalent if and only if $\alpha = \beta$.

Five-dimensional, solvable subalgebras of A_2

The unique 5-dimensional solvable subalgebra of A_2, up to inner automorphism, is a Borel subalgebra:

$$B \equiv \langle x_1, x_2, x_3, h_1, h_2 \rangle.$$

Dim.	Semisimple	Levi decomposable	Solvable
1	None	None	$J^1, J^2, J^3, J^{4,\alpha}$ where $J^{4,\alpha} \sim J^{4,\beta}$ if and only if $\alpha = 1-\beta, \alpha = \beta$, $\alpha = (-1+\alpha)\beta, \alpha = -(-1+\alpha)(-1+\beta)$, $\alpha\beta = 1$, or $-\alpha(-1+\beta) = 1$.
2	None	None	$K_1^i, 1 \leqslant i \leqslant 5, K_2^1, K_2^2, K_2^3, K_2^{4,\alpha}$ where $K_2^{4,\alpha} \sim K_2^{4,\beta}$ if and only if $\alpha = \beta$.
3	A_1^1, A_1^2	None	$L_2^1, L_2^2, L^1_{3,-\frac{1}{4}}, L^2_{3,-\frac{1}{4}}, L^1_{3,-\frac{2}{9}}, L^1_{3,0}, L_4^1, L_4^2, L_5^1$, $L^{1,\alpha}_{3,-\frac{(2\alpha-1)(\alpha-2)}{9(\alpha-1)^2}}, L^{2,\alpha}_{3,-\frac{(2\alpha-1)(\alpha-2)}{9(\alpha-1)^2}}, \alpha \neq \pm 1$, where, for $i = 1, 2$, $L^{i,\alpha}_{3,-\frac{(2\alpha-1)(\alpha-2)}{9(\alpha-1)^2}} \sim L^{i,\beta}_{3,-\frac{(2\beta-1)(\beta-2)}{9(\beta-1)^2}}$ if and only if $\alpha = \beta$ or $\alpha\beta = 1$.
4	None	$(A_1 \oplus J)^1$	$M_8^1, M_8^1, M_{12}^1, M_{13,0}^2, M_{13,2}^2, M_{14}^1, M^{1,\alpha}_{13,\frac{(2\alpha-1)(\alpha-2)}{(\alpha+1)^2}}, \alpha \neq \pm 1$, where $M^{1,\alpha}_{13,\frac{(2\alpha-1)(\alpha-2)}{(\alpha+1)^2}} \sim M^{1,\beta}_{13,\frac{(2\beta-1)(\beta-2)}{(\beta+1)^2}}$ if and only if $\alpha = \beta$.
5	None	$(A_1 \Subset_{\varphi_1} K_1)^1$, $(A_1 \Subset_{\varphi_1} K_1)^2$	B
6	None	$(A_1 \Subset_{\varphi_2} L_2)^1$, $(A_1 \Subset_{\varphi_2} L_2)^2$	None

The GraviGUT algebra as a subalgebra of E_8?

- The Standard Model of particle physics, with gauge group $\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$, attempts to describe all particles and all forces, except gravity. Grand Unified Theories (GUT) attempt to unify the forces and particles of the Standard Model.
- GUTs: Georgi and Glashow's $\mathrm{SU}(5)$ theory, Georgi's $\mathrm{Spin}(10)$ theory, and the Pati-Salam model $\mathrm{SU}(2) \times \mathrm{SU}(2) \times \mathrm{SU}(4)$.
- Lisi (2011) attempted to construct a unification which includes gravity. In this construction, Lisi first embeds gravity and the standard model into $\mathfrak{spin}(11,3)$. He then embeds $\mathfrak{spin}(11,3)$ together with the positive chirality 64-dimensional $\mathfrak{spin}(11,3)$ irrep into the quaternionic real form of E_8. Lisi refers to the embedded Lie algebra as the GraviGUT algebra.

The GraviGUT algebra as a subalgebra of E_8?

- $\text{GraviGUT}_{\mathbb{C}} = \mathfrak{so}(14, \mathbb{C}) \Subset V$. V is a 64 dimensional irrep of $\mathfrak{so}(14, \mathbb{C})$.
- Lisi's construction implies the existence of an embedding:

$$\mathfrak{so}(14, \mathbb{C}) \Subset V \hookrightarrow E_8$$

The GraviGUT algebra as a subalgebra of E_8?

Theorem (D., and Repka (2014))

Consider a sum of complex vector spaces $\mathfrak{so}(14,\mathbb{C}) \oplus V$, where V is a 64-dimensional space. Suppose a Lie bracket is defined which gives the usual structure to the $\mathfrak{so}(14,\mathbb{C})$ subspace and in such a way that the brackets $[X, v]$, for $X \in \mathfrak{so}(14,\mathbb{C})$ and $v \in V$, define an action of $\mathfrak{so}(14,\mathbb{C})$ on V under which V becomes an irreducible $\mathfrak{so}(14,\mathbb{C})$-module. Then the only way to extend this bracket to make $\mathfrak{so}(14,\mathbb{C}) \Subset V$ into a Lie algebra is to put the abelian structure on V. In particular, the only possible definition of the (complexified) GraviGUT algebra as a Lie algebra is $\mathfrak{so}(14,\mathbb{C}) \Subset V$, where V is a 64-dimensional abelian ideal which is irreducible under the action of $\mathfrak{so}(14,\mathbb{C})$.

The GraviGUT algebra as a subalgebra of E_8?

Proof.
Let V be a 64-dimensional $\mathfrak{so}(14,\mathbb{C})$-irrep. Then $V \cong V(\lambda_6)$ or $V \cong V(\lambda_7)$. Consider the tensor product decompositions:

$$V(\lambda_6) \otimes V(\lambda_6) \cong V(2\lambda_6) \oplus V(\lambda_5) \oplus V(\lambda_3) \oplus V(\lambda_1)$$
$$V(\lambda_7) \otimes V(\lambda_7) \cong V(2\lambda_7) \oplus V(\lambda_5) \oplus V(\lambda_3) \oplus V(\lambda_1)$$

Since $V(\lambda_6)$ does not occur in the tensor product decomposition of $V(\lambda_6) \otimes V(\lambda_6)$, $V(\lambda_7)$ does not occur in the tensor product decomposition of $V(\lambda_7) \otimes V(\lambda_7)$, and neither decomposition contains a 91-dimensional irrep, we cannot have a nontrivial product $V \otimes V \to V$ or $V \otimes V \to \mathfrak{so}(14,\mathbb{C})$. Hence, we cannot have a nontrivial product $V \times V \to V$ or $V \times V \to \mathfrak{so}(14,\mathbb{C})$ or $V \times V \to (\mathfrak{so}(14,\mathbb{C}) \Subset V)$. The only possible definition of a Lie algebra structure is to make V an abelian subalgebra. $\square$

The GraviGUT algebra as a subalgebra of E_8?

Corollary (D., and Repka (2014))

The GraviGUT algebra cannot be embedded into the quaternionic real form of E_8, or any other real form of E_8.

Proof.

The maximal dimension of an abelian subalgebra of E_8 is 36 [Malcev, 1951]. This implies that V cannot be a subalgebra of E_8, and that the complexified GraviGUT algebra cannot be embedded into E_8. The result follows. □

The GraviGUT algebra as a subalgebra of E_8?

- The Extended GraviGUT algebra is a Levi decomposable algebra $\mathfrak{so}(14,\mathbb{C}) \ltimes W$, where W, as an $\mathfrak{so}(14,\mathbb{C})$-rep, is the direct sum of a 64-dimensional $\mathfrak{so}(14,\mathbb{C})$-irrep with the standard $\mathfrak{so}(14,\mathbb{C})$ representation of dimension 14.
- In the same article mentioned above [Douglas, and Repka (2014a)] , we showed that the Extended GraviGUT algebra can be embedded into E_8.
- We classified these embeddings $\mathfrak{so}(14,\mathbb{C}) \ltimes W \hookrightarrow E_8$, up to inner automorphism.

The End.